TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Contents

List of HOTAS Tables

Chapter 1

HOTAS Fundamentals

The F-16's HOTAS (Hands On Throttle And Stick) controls enable the pilot to execute critical flight, sensor, and weapon functions without removing hands from the throttle and side-stick controller. Within this architecture, three switches — the Display Management Switch (DMS), Target Management Switch (TMS), and Countermeasures Management Switch (CMS) — control display selection, sensor management, and defensive systems.

For the complete HOTAS control reference, see *TO 1F-16CMAM-34-1-1*, Dash-34, sections `2.1.1.1.4 and 2.1.5`.

Two foundational concepts govern HOTAS switch behavior:

- **Sensor of Interest (SOI):** determines which display receives HOTAS inputs.
- **Master Modes (NAV, A-A, A-G):** define the available weapon systems and sensor configurations.

1.1 Sensor of Interest (SOI)

The F-16 cockpit contains three displays: the Head-Up Display (HUD) and two Multifunction Displays (MFD). The Sensor of Interest (SOI) determines which of these displays currently receives HOTAS inputs. At any moment, *only one* display can hold SOI designation. The active format on the display determines how the input is interpreted.

1.1.1 SOI Fundamentals

The SOI mechanism manages *where* HOTAS inputs are directed, not *what* those inputs do. The specific function of a HOTAS input — for example, whether TMS Up designates a target, breaks a track, or cycles through a mode — depends on delivery modes, current Master Mode, and active sensor format.

1.1.2 SOI Symbology

SOI designation is indicated by a distinct visual marker on each display:

- **HUD as SOI:** An asterisk symbol (*) appears in the upper left corner of the HUD, above the airspeed scale.
- **MFD as SOI:** A border outline is drawn around the edges of the MFD display, forming a box that distinguishes the SOI display from the non-SOI display.

- **NOT SOI indication:** The text `NOT SOI` appears in the center of any SOI-capable MFD format that is not currently designated as SOI. This indicator does not appear on MFD formats that cannot be designated as SOI.

1.1.3 SOI-Capable Displays

Not all displays can be designated as SOI, and the availability of SOI designation varies by Master Mode and MFD format.

Master Mode determines HUD eligibility for SOI designation. MFD formats can be designated as SOI in all Master Modes. The HUD, however, can be designated as SOI only in NAV and A-G Master Modes.

MFD format determines SOI eligibility *within* the MFD. Sensor formats that accept targeting and cursor control inputs — such as FCR, TGP, and WPN — can be designated as SOI. Informational and configuration formats, however — such as DTE, TCN, and FLCS — cannot be designated as SOI.

For SOI availability by Master Mode and MFD format, see Section **??**.

1.2 Master Modes

Master Mode is the highest-level configuration state of the F-16 avionics suite. Cockpit displays, sensor modes, and weapon options are configured by the active Master Mode. SOI availability and switch functions are also determined by Master Mode.

1.2.1 Primary and Override Master Modes

Three primary Master Modes (NAV, A-A, and A-G) and two override modes (DGFT and MSL OVRD) are available in the F-16:

- **Navigation (NAV):** The default Master Mode, active when no other mode is selected. Both air-to-air and air-to-ground sensor modes are available in NAV. Navigation symbology is displayed on the HUD.

- **Air-to-Air (A-A):** Selected via the A-A button on the ICP. Only air-to-air sensor modes are available in A-A. Air-to-air weapon selection and engagement symbology are displayed on the HUD.

- **Air-to-Ground (A-G):** Selected via the A-G button on the ICP. Both air-to-air and air-to-ground sensor modes are available in A-G. Air-to-ground weapon delivery symbology is displayed on the HUD.

- **Dogfight (DGFT) and Missile Override (MSL OVRD):** Selected via the Dogfight/Missile Override switch on the throttle grip — DGFT outboard, MSL OVRD inboard. Only air-to-air sensor modes are available in both modes. DGFT and MSL OVRD take precedence over all other Master Modes except Emergency Jettison.

Default sensor, display, and weapon options are configured by the active Master Mode. These defaults are programmable via the Data Transfer Cartridge (DTC) or adjustable manually in-flight. For complete Master Mode details, see Dash-34 § 2.1.1.2.1.